

PROJECT REPORT ON WEB PORTAL DEVELOPMENT FOR "CCL KE LAL · CCL KI LAADLI"

Submitted in partial fulfillment of the requirements for the Internship Program

TABLE OF CONTENTS

1. [Acknowledgments](#1-acknowledgments)
 2. [Abstract](#2-abstract)
 3. [Introduction](#3-introduction)
 4. [Company Profile: Central Coalfields Limited](#4-company-profile-central-coalfields-limited)
 5. [Problem Statement & Objectives](#5-problem-statement--objectives)
 6. [System Requirements & Technology Stack](#6-system-requirements--technology-stack)
 7. [System Architecture & File Structure](#7-system-architecture--file-structure)
 8. [Frontend Design & Implementation](#8-frontend-design--implementation)
 9. [Backend Development & PHP Logic](#9-backend-development--php-logic)
 10. [Database Design & Schema](#10-database-design--schema)
 11. [Module Descriptions (Features)](#11-module-descriptions-features)
 12. [Security & Session Management](#12-security--session-management)
 13. [Software Development Life Cycle (SDLC)](#13-software-development-life-cycle-sdlc)
 14. [Challenges Encountered & Solutions](#14-challenges-encountered--solutions)
 15. [Future Enhancements & Scope](#15-future-enhancements--scope)
 16. [Conclusion](#16-conclusion)
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1. Acknowledgments

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I extend my sincere thanks to my project guides and mentors within the CCL IT and CSR departments, whose guidance, technical expertise, and continuous feedback were instrumental in the successful completion and modernization of this web application.

2. Abstract

In the modern digital era, educational initiatives require robust digital infrastructures to streamline operations, facilitate information dissemination, and manage student data effectively. This project report details the end-to-end development, refactoring, and modernization of a web portal for **CCL Ke Lal · CCL Ki Laadli**, a flagship Corporate Social Responsibility (CSR) initiative by Central Coalfields Limited.

The primary objective was to transition a static, legacy web presence into a dynamic, stateful, and visually premium Full-Stack Web Application. Leveraging the LAMP/WAMP architecture (Windows, Apache, MySQL, PHP), the project encompasses a complete frontend overhaul utilizing CSS3 Glassmorphism and responsive grids, paired with a secure PHP backend. Key features developed include an automated student registration pipeline, secure session-based user authentication, an interactive student dashboard, and a seamless quiz-taking interface with automated score recording directly into a relational database system.

3. Introduction

Corporate Social Responsibility (CSR) is a crucial aspect of modern corporate governance. Central Coalfields Limited's initiative, ***CCL Ke Lal · CCL Ki Laadli***, provides free boarding, lodging, and highly specialized educational coaching to meritorious students preparing for competitive engineering entrance examinations like the IIT-JEE.

As the program scales, relying on manual data entry and disjointed static web pages becomes inefficient. The need for a centralized web portal became evident to handle online applications, provide verified information to parents/guardians, and act as a digital hub for enrolled students. This internship project focused on designing this exact digital hub, ensuring it is highly scalable, user-friendly, and secure.

4. Company Profile: Central Coalfields Limited

Central Coalfields Limited (CCL) is a Category-I Mini-Ratna Company and a subsidiary of Coal India Limited (CIL), the largest coal-producing company in the world. Formed on November 1, 1975, CCL operates primarily in the state of Jharkhand, managing numerous coal mines, washeries, and central workshops.

Beyond its core industrial operations, CCL is deeply committed to the socio-economic upliftment of the communities residing in its command areas. The CSR initiatives undertaken by CCL span healthcare, infrastructure development, and education. The ***CCL Ke Lal · CCL Ki Laadli*** scheme is the crown jewel of their educational outreach, boasting a stellar track record of transforming the lives of underprivileged students by granting them access to premier engineering institutions.

5. Problem Statement & Objectives

5.1 Problem Statement

The pre-existing digital footprint of the scheme consisted of disconnected HTML pages with archaic styling (relying on deprecated tags like `` and inline styles). More critically, the backend functionality for registration and login was non-functional due to misconfigurations in the PHP architecture and a lack of a proper database schema, resulting in raw source code being downloaded by the client browser rather than being executed by the server.

5.2 Objectives

1. **Modernize the UI/UX:** Refactor all frontend code to meet modern web standards (HTML5/CSS3) and implement a premium, fully responsive design system.
 2. **Develop a Dynamic Backend:** Implement server-side logic using PHP to handle HTTP POST requests securely.
 3. **Database Integration:** Design and provision a MySQL database to persist student application data, login credentials, and academic quiz records.
 4. **Authentication System:** Build a secure login mechanism utilizing PHP Sessions.
 5. **Interactive Dashboard:** Create a private student area where enrolled students can take online quizzes and track their historical performance.
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6. System Requirements & Technology Stack

6.1 Hardware Requirements

- **Processor:** Intel Core i3 / AMD Ryzen 3 (or equivalent) minimum.
- **RAM:** 4GB minimum (8GB recommended for running Apache/MySQL servers locally alongside an IDE).
- **Storage:** 500MB free disk space for XAMPP installation and project files.

6.2 Software & Technology Stack (WAMP Environment)

The project utilizes the classic client-server model heavily reliant on the WAMP stack:

- **Operating System:** Windows 10/11
- **Web Server (Apache):** Handles incoming HTTP/HTTPS requests and serves static assets and parsed PHP scripts.

- **Database (MySQL / MariaDB):** A relational database management system (RDBMS) utilizing the InnoDB storage engine for ACID compliance.
 - **Backend Language (PHP 8.x):** Hypertext Preprocessor, a server-side scripting language used to connect the UI with the database.
 - **Frontend Languages:**
 - **HTML5:** For semantic Document Object Model (DOM) structuring.
 - **CSS3:** For styling, Flexbox/Grid layouts, and CSS variables.
 - **Vanilla JavaScript (ES6):** For client-side interactivity and carousel logic.
 - **IDE / Environment:** Visual Studio Code / Antigravity IDE.
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7. System Architecture & File Structure

7.1 Architecture Pattern

The application follows a standard **Client-Server Architecture**.

1. The **Client** (Web Browser) sends an HTTP GET/POST request.
2. The **Apache Server** intercepts the request. If the requested file is `.php``, it routes it to the PHP preprocessor.
3. The **PHP Engine** executes the logic, communicates with the **MySQL Database** (if querying/inserting data is required), and compiles the output into standard HTML.
4. The **Server** sends the compiled HTML back to the Client as an HTTP Response.

7.2 Directory Structure

```text

/htdocs/CCL-Internship-Project-main/Home/

■■■ index.html # Landing Page / Hero Section

■■■ aboutccl.html # Information regarding CCL history

■■■ aboutscheme.html # Details regarding the CSR scheme

■■■ aboutschool.html # Information on DAV Public School

■■■ contact.html # Contact & Social links

■■■ global-theme.css # Centralized UI Design System (CSS3)

■■■ form1.php # Registration Form & Backend Insertion Logic

■■■ form2.php # Login Form & Authentication Logic

■■■ register.php # Secured Student Dashboard & Quiz History

■■■ Quiz1.php / Quiz2.php # Interactive Quiz Modules

■■■ dashboard.php # Quiz Submission Handler

■■■ connection1.php # Database Connection Middleware

■■■ logout.php # Session Destruction Script

■■■ database\_setup.sql # DDL Script for Database Provisioning

■■■ [Image Assets] # Various .jpg and .png files for UI

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## 8. Frontend Design & Implementation

### 8.1 The Global Theme (`global-theme.css`)

To achieve a highly professional and cohesive aesthetic, a singular global stylesheet was engineered. This adhered to the DRY (Don't Repeat Yourself) principle of software engineering.

- **CSS Variables (Custom Properties):** Defined at the ``:root` level to maintain consistent primary (`#4F46E5`), secondary (`#EC4899`), and background colors.
- **Typography:** Integrated Google Fonts. `Inter` was utilized for body text due to its excellent legibility on digital screens, while `Poppins` was reserved for high-impact headings.

### 8.2 Glassmorphism & UI Cards

The outdated layout utilizing horizontal rules (``) was completely replaced with a modern "Glassmorphism" UI.

- **Implementation:** Content sections were encapsulated in ``. These cards utilized a semi-transparent background (`rgba(255, 255, 255, 0.85)`), soft border radiuses (16px - 20px), and CSS filters (`backdrop-filter: blur(16px)`) to create a frosted glass effect over the underlying radial-gradient body background.
- **Micro-interactions:** CSS transitions were applied to hover states (`transform: translateY(-5px)`) to provide tactile, responsive feedback to the user.

### 8.3 Responsive Web Design (RWD)

The UI was built to be responsive across desktop and mobile devices. Rather than relying on heavy CSS frameworks like Bootstrap, native CSS Grid and Flexbox were utilized.

- Example: ``.grid-2 { display: grid; grid-template-columns: repeat(auto-fit, minmax(300px, 1fr)); }` ensures that content cards automatically reflow from a multi-column layout on desktops to a single-column layout on mobile devices.
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## 9. Backend Development & PHP Logic

### 9.1 Database Connectivity (`connection1.php`)

A middleware script was written to establish a persistent connection between the PHP runtime and the MySQL server using the `mysqli\_connect()` function. It passes the `\$servername`, `\$username`, `\$password`, and `\$dbname` parameters and returns a connection object (`\$conn`) used globally across the application.

### 9.2 Registration Pipeline (`form1.php`)

- **Data Ingestion:** The HTML form collects 15 unique data points (Name, DOB, Category, Board, etc.) utilizing the `POST` method.
- **Payload Extraction:** PHP extracts this payload via the `\$\_POST` superglobal array. Null coalescing operators (`??`) were used to prevent undefined index errors.
- **Query Execution:** An `INSERT INTO form1 (...) VALUES (...)` SQL query is dynamically constructed and executed via `mysqli\_query()`.
- **Dynamic Feedback:** Upon successful execution, a success alert box is injected into the DOM; otherwise, an error state is displayed.

### 9.3 Quiz Submission Logic (`dashboard.php`)

When a student completes a quiz via the JS-driven interface in `Quiz1.php`, the final score is passed via a GET request (`?id=score`) to `dashboard.php`.

- **Data Sanitization:** Before insertion, `mysqli\_real\_escape\_string()` is applied to the user ID and score to sanitize inputs and prevent basic SQL injection vectors.
- **Persistence:** The score and the active student's email are inserted into the `result` database table.

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## 10. Database Design & Schema

The relational database (`registrationform`) was provisioned using a custom Data Definition Language (DDL) script (`database\_setup.sql`).

### 10.1 Table 1: `form1` (Student Records)

This table acts as the master record for all applicants.

- `id`: INT, Primary Key, Auto Increment.
- `fname`, `lname`: VARCHAR(50), stores applicant's name.
- `dob`: DATE, serves as the authentication password.
- `student\_email`: VARCHAR(100), UNIQUE constraint applied to prevent duplicate registrations and act as the User ID.
- `category`, `boardname`, `per`, `ccl`: Stores academic and demographic data for eligibility filtering.

- ``created_at``: `TIMESTAMP`, defaults to ``CURRENT_TIMESTAMP``.

## 10.2 Table 2: ``result`` (Quiz Analytics)

This table records the performance of students in internal assessments.

- ``id``: `INT`, Primary Key, Auto Increment.
  - ``student_email``: `VARCHAR(100)`, acts as a Foreign Key linking to the ``form1`` table.
  - ``test``: `VARCHAR(10)`, stores the numerical score achieved.
  - ``submitted_at``: `TIMESTAMP`, records the exact date and time the quiz was completed.
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## 11. Module Descriptions (Features)

1. **Information Dissemination Module:** Static but visually engaging pages (``aboutccl.html``, ``aboutscheme.html``, ``aboutschoool.html``) that detail the history, selection methodology, syllabus, and contact information.
  2. **Registration Module:** A comprehensive, multi-field form equipped with HTML5 validation (required fields, email formatting, date pickers) that routes data to the MySQL backend.
  3. **Authentication Module:** A login portal verifying credentials against the ``form1`` table and initiating server sessions.
  4. **Student Dashboard Module:** A secured, private route (``register.php``) accessible only to logged-in users. Features include:
    - Dynamic welcome greeting using the active Session ID.
    - An automated table that queries the ``result`` table and renders the student's historical quiz scores and timestamps.
  5. **Interactive Quiz Module:** Client-side JavaScript quizzes featuring pagination, dynamic scoring, and automated redirection to the backend submission handler.
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## 12. Security & Session Management

### 12.1 Stateful Sessions

Because the HTTP protocol is inherently stateless, PHP Sessions were utilized to maintain authentication state.

- **Login:** Upon verified credentials, ``session_start()`` is invoked. ``$_SESSION['flag'] = 1`` and ``$_SESSION['userid'] = $email`` are set in the server memory.
- **Route Protection:** The dashboard (``register.php``) begins by checking ``if(!isset($_SESSION['flag']) || $_SESSION['flag'] != 1)``. If unauthorized, it executes an HTTP 302 Redirect (``header('Location: form2.php')``) kicking the user back to the login page.

- **Logout:** `logout.php` safely terminates the connection by invoking `session_unset()` (clearing variables) and `session_destroy()` (deleting the session file from the server), before redirecting to the login screen.

## 12.2 Data Integrity

- **Unique Constraints:** The database utilizes a `UNIQUE` index on the `student_email` column, structurally preventing malicious or accidental duplicate account creation.
  - **Sanitization:** `mysqli_real_escape_string()` was implemented during database insertions (e.g., the quiz submission) to neutralize special characters that could alter SQL logic, providing a baseline defense against SQL Injection (SQLi).
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## 13. Software Development Life Cycle (SDLC)

The project followed an **Agile-Iterative Methodology**:

1. **Requirement Analysis:** Understanding the legacy code, identifying the bugs (static vs. dynamic server issues), and charting the goals of the CSR scheme.
  2. **Design Phase:** Architecting the database schema, selecting the color palette, and mapping out the user journey.
  3. **Development (Iteration 1):** Overhauling the frontend HTML/CSS to implement the glassmorphic design system.
  4. **Development (Iteration 2):** Developing the PHP backend, writing the SQL queries, and integrating the database connection.
  5. **Testing & Debugging:** Running the application on XAMPP, fixing case-sensitivity errors in file inclusions, resolving port conflicts, and verifying session integrity.
  6. **Deployment & Review:** Finalizing the UI/UX, removing redundant elements (e.g., legacy "Apply Now" buttons), and packaging the SQL setup script.
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## 14. Challenges Encountered & Solutions

### 14.1 Static vs. Dynamic Server Conflict

**Challenge:** The initial environment was running on a basic Python static HTTP server (`python -m http.server`), causing PHP files to download as raw text rather than executing.

**Solution:** Migrated the entire workspace into the `htdocs` directory of an Apache HTTP Server via XAMPP. This enabled the PHP preprocessor to compile server-side logic before serving it to the client browser.

### 14.2 Outdated Legacy UI



**Challenge:** The provided HTML lacked responsiveness, making it unreadable on mobile devices, and relied on deprecated alignment tags.

**Solution:** Completely refactored the DOM. Wrote a `global-theme.css` utilizing modern Flexbox/Grid layouts, fluid typography, and standardized padding/margins, resulting in a premium corporate aesthetic.

## 14.3 Unlinked Quiz Results

**Challenge:** The quizzes calculated scores via JavaScript but failed to persist that data to the student's profile, limiting administrative tracking.

**Solution:** Rewrote the quiz submission logic to pass data to a PHP handler. Altered the database schema to include a `result` table, and modified the student dashboard to actively query and display these results associated with the active session's User ID.

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## 15. Future Enhancements & Scope

While the current portal successfully meets the operational requirements of the CSR initiative, several enhancements can be integrated in future iterations:

1. **Password Hashing:** Currently, the system uses the student's Date of Birth (DOB) in plaintext as a password. Future versions must implement the `password\_hash()` (Bcrypt/Argon2) function for cryptographic security.
2. **Prepared Statements:** Transitioning from concatenated `mysqli\_query` statements to PDO (PHP Data Objects) or `mysqli\_prepare()` to provide robust, enterprise-grade protection against SQL Injection.
3. **Admin Dashboard:** Creating a separate portal (`admin.php`) where CCL executives can view aggregated analytics, download student applicant data in CSV format, and manage quiz content dynamically without altering source code.
4. **Email Integration:** Implementing PHPMailer or SMTP protocols to automatically send confirmation emails and admit cards to students upon successful registration.

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## 16. Conclusion

The development of the web portal for the **CCL Ke Lal · CCL Ki Laadli** initiative proved to be a highly successful and educational endeavor. The project evolved from a fragmented, static prototype into a cohesive, secure, and visually stunning Full-Stack Web Application.

By implementing modern UI/UX principles alongside a robust LAMP/WAMP architecture, the portal now provides a frictionless experience for underprivileged, meritorious students seeking to apply for this transformative CSR program. Furthermore, the robust backend ensures that CCL administrators can reliably capture, store, and process applicant data. This internship project not only fulfilled the technical requirements of the organization but also provided profound, hands-on experience in modern web engineering, database architecture, and system deployment.